

TeleFormation: Join Socially Interacting with Comfortable for Shared Virtual Environment using F-Formation

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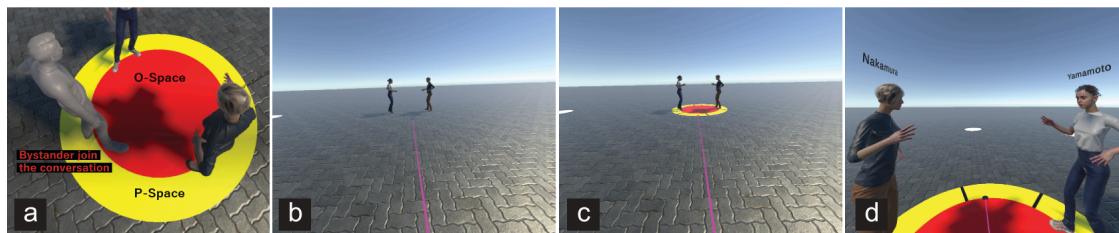
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Figure 1: TeleFormation investigates the plausible comfort join socially conversation in a virtual environment (a) through the comparison between (b) without and (c) with the personal space projection. We believe that using personal space could prevent socially inappropriate intrusions into the joint conversation (d).

ABSTRACT

Maintaining appropriate interpersonal distance is essential for user comfort and privacy protection in a socially aware context. However, existing approaches provide limited support for respecting personal space, leaving users vulnerable to unintentional or socially inappropriate intrusions. This paper addresses this issue by investigating an alternative method to enhance comfort when joining a pair of conversing users by leveraging the concept of F-formation. Our results provide a comfortable approach to teleportation into a target group of conversation through 6- and 12-sector visualizations in the F-formation method using P-space.

Index Terms: Social VR, Virtual Space, Teleportation.

1 INTRODUCTION

Teleportation is a common locomotion technique in Virtual Reality (VR) that is widely adopted in many interactions, such as to allow instant transport of a user from one location to another in a virtual environment (VE) [4], and to allow users to approach others from a distance to facilitate a conversation or to observe the discussion. In the latter, teleportation could enhance social interaction. However, a significant issue arises when users teleport to the conversation without selecting an appropriate approach distance, leading to violations of personal space and hindering comfortable social interactions.

Previous work proposed negotiated teleportation for multi-user conversational scenarios [5] or relied on Proxemics theory [1], which examines how individuals utilize the space surrounding them. Nevertheless, understanding the nature of such an approach toward multiple individuals remains limited [3]. A primary limitation of traditional proxemics theory is its heavy reliance on physical distance while neglecting the importance of body orientation and mutual gaze. Because proxemics focuses primarily on proximity, it may incorrectly categorize individuals as being in a 'close' relationship even when they are standing back-to-back. Consequently,

this method fails to account for the directional nature of social interaction, which is essential for analyzing conversational dynamics.

This paper investigates an alternative method to enhance comfort when joining a pair of conversing users by leveraging the concept of F-Formation [2] called *TeleFormation*. Utilizing F-Formation, we determine the specific spatial arrangements of users during physical encounters involving group interaction (Figure 1(a)). As shown in Figure 1(b-d), we compare our proposed 6- and 12-sectors personal space with the traditional non-space division method.

2 TELEFORMATION

TeleFormation is an interface for visualizing a shared VE that helps users enter the conversation. Assuming the conversation in a shared VE is usually in a facing formation (F-formation), our method uses personal-space-aware locomotion to display possible teleportation locations that do not violate personal space during group conversation. While we believe that the formation of guides helps users join the conversation in a social context, it is unknown whether the number of sectors affected could affect comfort and usability. As shown in Figure 2, we investigate three types of visualization: (1) without subdivision of the P-space (NS), (2) with the P-space divided into six sectors (6S), and (3) with the P-space divided into twelve sectors (12S). Comparing them with the no-visualization (NV) as the traditional method.

3 PRELIMINARY EXPERIMENT

Five participants (avg. 20.3 years old, all males) were recruited from the local university. The experimental system was developed

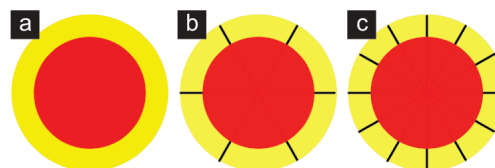


Figure 2: Formations visualization uses in the experiment: (a) without subdivision (NS), (b) six-sectors (6S), and (c) twelve-sectors (12S).

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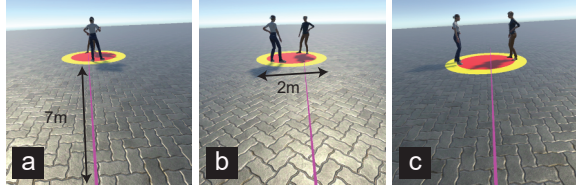


Figure 3: Start position of each condition: (a) 0°, (b) 45°, and (c) 90°, and the start position is at 7m away.

using Unity (ver.6000.1.8f1) and a Head-Mounted Display (Pico 4 Ultra). To reduce the number of participants in the social conversation, we use a virtual agent from Mixamo. Thus, only participants who take the role of the joint conversation will participate in this preliminary experiment.

The experiment employed a 4×3 within-subjects design. The independent variables consisted of four presentation methods, i.e., a no visual feedback and three types of F-formation visualizations, and three teleportation starting positions. These starting positions were located at 7 m away on the 0°, 45°, and 90° to a circle with a radius of 2 m, centered on the origin of the O-space (Figure 3).

Participants were instructed to teleport to an appropriate position upon the appearance of a "Start" prompt on the screen. They can take their time to consider the appropriate position to take in the conversation without a time limit. After joining the conversation, they were asked to verbally identify the name displayed on a tag above the head of each agent.

For subjective evaluation, usability was assessed using a scale adapted from the USE Questionnaire. We also utilized the PSIAL framework to evaluate distance comfort and assess the perceived appropriateness of the interpersonal distance maintained upon joining the group. For objective evaluation, Task Completion Time (TCT) was measured as the time from the appearance of the "Start" prompt until the participant verbally identified the name displayed on the tag above the agent.

4 RESULTS

Comfortable As shown in Figure 4, The results on NV, NS, 6S, and 12S in average at 0° are 4.07 (SD 0.37), 4.67 (SD 1.05), 4.40 (SD 1.16), and 5.30 (SD 1.51), respectively. At 45°, 4.67 (SD 1.07), 5.23 (SD 1.05), 5.80 (SD 0.61), and 4.80 (SD 1.19), respectively. At 90°, 5.60 (SD 0.45), 6.07 (SD 0.47), 5.80 (SD 0.78), and 5.80 (SD 0.82), respectively. The 12S method was most effective when teleporting from 0°, whereas the 6S method demonstrated superior performance at 45°. At an angle of 90°, no significant differences in scores were observed. Overall, the results indicate that smaller teleportation starting angles tend to diminish user comfort.

Usability As shown in Figure 5, the results on NV, NS, 6S, and 12S in average at 0° are 3.47 (SD 0.38), 3.60 (SD 1.38), 4.40 (SD 0.84), and 4.33 (SD 0.98), respectively. At 45°, 4.80 (SD 1.02), 4.33 (SD 0.82), 5.73 (SD 0.44), and 4.80 (SD 1.37), respectively. At 90°, 5.13 (SD 1.02), 5.93 (SD 0.64), 5.53 (SD 0.73), and 5.40 (SD 0.44), respectively. Our results found that both 6S and 12S were effective at 0°, while the NV resulted in markedly lower scores. At 45°, 6S were found to be the most effective, whereas 12S exhibited a wide variance in scores. Overall, usability scores for the 0° starting position were lower than those for other conditions.

TCT The time required to complete the tasks are 9.26s (SD 1.96), 9.49s (SD 3.33), 9.43 (SD 3.28), and 10.70s (SD 4.24) for NV, NS, 6S, and 12S, respectively. The 12S method required significantly more time than the other methods.

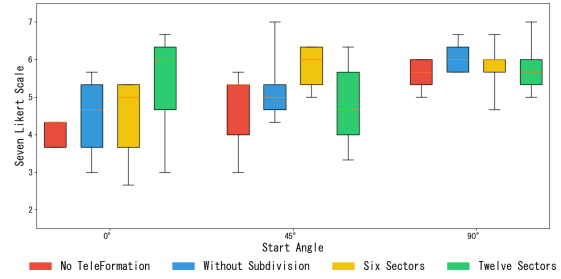


Figure 4: Comfortable score for each angle (7 = very comfortable).

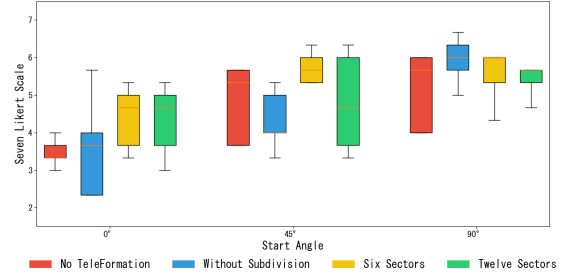


Figure 5: Usability score for each angle (7 = very useful).

5 DISCUSSION

Overall, experimental results demonstrated that segmented visualizations enhance both comfort and usability. Particularly, when teleporting from a 0° angle where the participants in the conversation are likely enclosed to each other. Conversely, increasing the number of subdivisions increases decision time. The participants seem to have difficulty deciding where to teleport for the group conversation. The twelve-sector method showed a significant increase in decision length compared to other conditions. Consequently, the six sectors method can be considered the most balanced approach, offering an optimal trade-off between user experience and efficiency.

6 CONCLUSION

This paper investigated a method to enhance comfort when joining a pair of conversing users by leveraging the concept of F-formation. Our results provide a comfortable approach to teleportation into a target group of conversation through 6- and 12-sector visualizations in the F-formation method using P-space. Future work should address these limitations by increasing the number of participants in the conversation and investigating other types of F-formations.

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